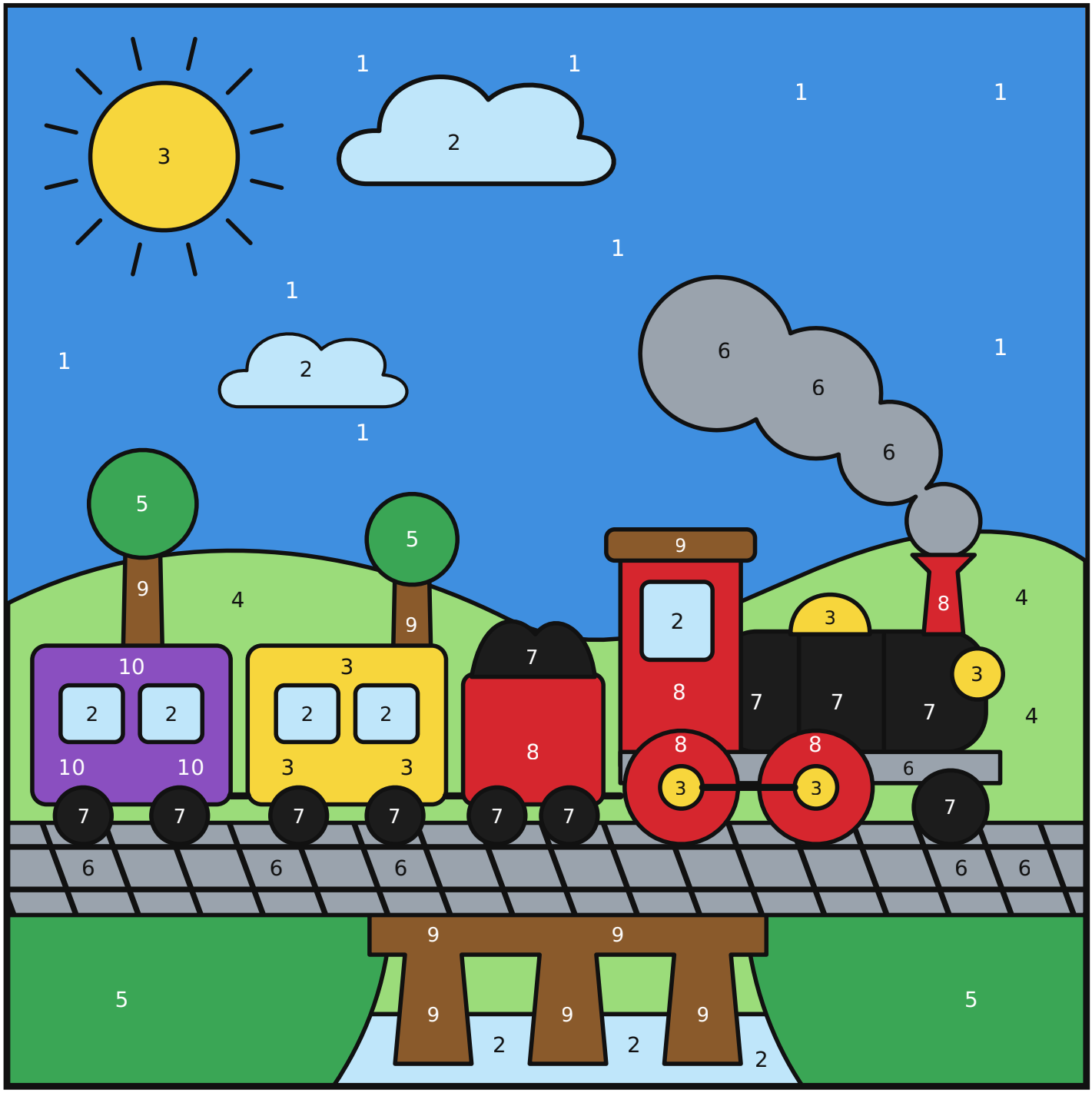
Name: **ANSWER KEY** _____ Date: _____

Angles & Parallel Lines: Color By Number



Color Code:

8 Red 90°	**5** Green 70°	**2** Light Blue 92°	**6** Gray 123°	**1** Blue 55°
3 Yellow 65°	**4** Light Green 56°	**10** Purple 720°	**7** Black 82°	**9** Brown 68°

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Angles & Parallel Lines: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
Parallel lines are cut by a transversal. Alternate interior angles measure (3x + 10)° and (5x − 20)°. Find the measure of each angle.

3x + 10 = 5x − 20, so x = 15
3(15) + 10
55°

1. Color: **Blue**

2.
Parallel lines are cut by a transversal. Same-side interior angles measure (2x + 20)° and (3x − 10)°. Find the larger angle.

5x + 10 = 180, so x = 34
angles: 88° and 92°
92°

2. Color: **Light Blue**

3.
Parallel lines are cut by a transversal. Corresponding angles measure (4x − 15)° and (2x + 25)°. Find the measure of each angle.

4x − 15 = 2x + 25, so x = 20
4(20) − 15
65°

3. Color: **Yellow**

4.
Parallel lines are cut by a transversal. Alternate exterior angles measure (6x − 4)° and (4x + 16)°. Find the measure of each angle.

6x − 4 = 4x + 16, so x = 10
6(10) − 4
56°

4. Color: **Light Green**

5.
The angles of a triangle measure 2x°, (3x + 5)° and (4x − 14)°. Find the largest angle.

9x − 9 = 180, so x = 21
angles: 42°, 68°, 70°
70°

5. Color: **Green**

6.
An exterior angle of a triangle measures (5x + 3)°. The two remote interior angles measure (2x + 30)° and (x + 21)°. Find the exterior angle.

5x + 3 = (2x + 30) + (x + 21)
2x = 48, so x = 24
5(24) + 3
123°

6. Color: **Gray**

7.
Two vertical angles measure (7x − 9)° and (5x + 17)°. Find the measure of each angle.

7x − 9 = 5x + 17, so x = 13
7(13) − 9
82°

7. Color: **Black**

8.
Angles measuring (x + 40)° and (2x − 10)° are supplementary. Find the measure of the second angle.

3x + 30 = 180, so x = 50
2(50) − 10
90°

8. Color: **Red**

9.
The vertex angle of an isosceles triangle measures 44°. Find the measure of each base angle.

base angles are equal: 2b + 44 = 180
2b = 136
68°

9. Color: **Brown**

10.
Find the sum of the interior angles of a hexagon.

sum = (n − 2) · 180 with n = 6
4 · 180
720°

10. Color: **Purple**